

| <b>VI Semester</b> |                    |                                                                   |                     |
|--------------------|--------------------|-------------------------------------------------------------------|---------------------|
| <b>S. No</b>       | <b>Course Code</b> | <b>Course Title</b>                                               | <b>Credit Hours</b> |
| 1.                 | <b>AGRON-321</b>   | Dryland Agriculture /Rainfed Agriculture and Watershed Management | 2(1+1)              |
| 2.                 | <b>AGRON-322</b>   | Introductory Agro-forestry                                        | 2(1+1)              |
| 3.                 | <b>AECON-321</b>   | Agricultural Finance and Co-Operation                             | 2(1+1)              |
| 4.                 | <b>BIOCH-321</b>   | Essentials of Plant Biochemistry                                  | 3(2+1)              |
| 5.                 | <b>BT-321</b>      | Fundamentals of Agriculture Biotechnology                         | 3(2+1)              |
| 6.                 | <b>GPB-321</b>     | Crop Improvement ( <i>Rabi</i> crops)-II                          | 2(1+1)              |
| 7.                 | <b>GPB-322</b>     | Fundamentals of Seed Science and Technology                       | 2(1+1)              |
| 8.                 | <b>PPATH-321</b>   | Agricultural Microbiology and Phyto-remediation                   | 2(1+1)              |
| 9.                 | <b>STAT-321</b>    | Basic and Applied Agri-Statistics                                 | 3(2+1)              |
| <b>TOTAL</b>       |                    |                                                                   | <b>21(12+9)</b>     |

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| AGR0N-321 | Dryland Agriculture /Rainfed Agriculture and Watershed Management | 2(1+1) |
|-----------|-------------------------------------------------------------------|--------|

### Objectives

1. To learn about characteristics and conditions of dryland/rainfed agriculture
2. To gain knowledge about drought and its mitigation
3. To impart knowledge on water harvesting and watershed management

### Theory

Dryland/Rainfed agriculture: Introduction, types and characteristics; History of dry land/ rainfed agriculture in India; Problems and prospects of dry land/rainfed agriculture in India; Soil and climatic conditions prevalent in dry land/rainfed areas; Length of Growing Period (LGP) and Soil Moisture Availability (SMA) and its impact on crop and cropping system; Soil and water conservation techniques; Drought: types, effect of water deficit on physio- morphological characteristics of the plants; Crop adaptation and mitigation to drought; Water harvesting: importance, its techniques, Efficient utilization of water through soil and crop management practices; Crops and cropping systems in dry land/rainfed areas; Management of crops in dry land/rainfed areas; Contingent crop planning for aberrant weather conditions; Concept, history, objective, principles and components of watershed management, factors affecting watershed management. Log term rainfall analysis in relation to simple mathematical models and forecasting the weather abnormalities; Alternate land use system location; regional and crop specific dryland principles and practices for profitable and sustainable dryland farming and allied enterprises.

### Practical

Studies on climate classification, studies on rainfall pattern in rainfed areas of the country and pattern of onset and withdrawal of monsoons. Calculation of Length of Growing Period (LGP) and Soil Moisture Availability (SMA) Studies on cropping pattern of different rainfed areas in the country and demarcation of rainfed area on map of India. Interpretation of meteorological data and scheduling of supplemental irrigation on the basis of evapo-transpiration demand of crops. Critical analysis of rainfall and possible drought period in the country. Effective rainfall and its calculation. Studies on cultural practices for mitigating moisture stress including mechanical and agronomic measure. Soil moisture determination under different land situations, Importance of seed priming to mitigate drought. Assessment of meteorological drought. Characterization and delineation of model watershed. Seed treatment, viz., seed hardening and seed priming techniques for all the agricultural crops Field demonstration on soil and moisture conservation measures. Field demonstration on construction of water harvesting structures. Visit to rainfed research station/watershed.

**Lecture schedule: Theory**

| S.N. | Topic                                                                                                                             | No. of lectures |
|------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.   | Dryland/Rainfed agriculture: Introduction, types and characteristics; History of dry land/ rainfed agriculture in India           | 1               |
| 2.   | Problems and prospects of dry land/rainfed agriculture in India; Soil and climatic conditions prevalent in dry land/rainfed areas | 1               |
| 3.   | Length of Growing Period (LGP) and Soil Moisture Availability (SMA) and its impact on crop and cropping system                    | 1               |
| 4.   | Soil and water conservation techniques                                                                                            | 1               |
| 5.   | Drought: types, effect of water deficit on physio- morphological characteristics of the plants                                    | 1               |
| 6.   | Crop adaptation and mitigation to drought                                                                                         | 1               |
| 7.   | Water harvesting: importance, its techniques                                                                                      | 1               |
| 8.   | Efficient utilization of water through soil and crop management practices                                                         | 1               |
| 9.   | Crops and cropping systems in dry land/rainfed areas                                                                              | 1               |
| 10.  | Management of crops in dry land/rainfed areas                                                                                     | 1               |
| 11.  | Contingent crop planning for aberrant weather conditions                                                                          | 1               |
| 12.  | Concept, history, objective, principles and components of watershed management, factors affecting watershed management            | 2               |
| 13.  | Log term rainfall analysis in relation to simple mathematical models and forecasting the weather abnormalities                    | 1               |
| 14.  | Alternate land use system location                                                                                                | 1               |
| 15.  | regional and crop specific dryland principles and practices for profitable and sustainable dryland farming and allied enterprises | 1               |

**Lecture schedule: Practical**

| S.N. | Topic                                                                                                                 | No. of lectures |
|------|-----------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.   | Studies on climate classification                                                                                     | 1               |
| 2.   | Studies on rainfall pattern in rainfed areas of the country and pattern of onset and withdrawal of monsoons           | 1               |
| 3.   | Calculation of Length of Growing Period (LGP) and Soil Moisture Availability (SMA)                                    | 1               |
| 4.   | Studies on cropping pattern of different rainfed areas in the country and demarcation of rainfed area on map of India | 1               |

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| 5.  | Interpretation of meteorological data and scheduling of supplemental irrigation on the basis of evapo-transpiration demand of crops | 1 |
| 6.  | Critical analysis of rainfall and possible drought period in the country                                                            | 1 |
| 7.  | Effective rainfall and its calculation                                                                                              | 1 |
| 8.  | Studies on cultural practices for mitigating moisture stress including mechanical and agronomic measure                             | 1 |
| 9.  | Soil moisture determination under different land situations                                                                         | 1 |
| 10. | Importance of seed priming to mitigate drought                                                                                      | 1 |
| 11. | Assessment of meteorological drought                                                                                                | 1 |
| 12. | Characterization and delineation of model watershed                                                                                 | 1 |
| 13. | Seed treatment, viz., seed hardening and seed priming techniques for all the agricultural crops                                     | 1 |
| 14. | Field demonstration on soil and moisture conservation measures                                                                      | 1 |
| 15. | Field demonstration on construction of water harvesting structures                                                                  | 1 |
| 16. | Visit to rainfed research station/watershed                                                                                         | 1 |

#### **Suggested readings**

1. A.K. Srivastava and P.K. Tyagi. 2011. Practical Agricultural Meteorology. New Delhi Publishing Agency, New Delhi.
2. D. Lenka. 2006. Climate, Weather and Crops in India. Kalyani Publishers, New Delhi.
3. G.S.L.H.V. Prasad Rao. 2008. Agricultural Meteorology. Prentice Hall of India Pvt. Ltd., New Delhi.
4. H.S. Mavi and Graeme J. Tupper. 2005. Agrometeorology – Principles and applications of climate studies in agriculture. International Book Publishing Co., Lucknow.
5. H.S. Mavi. 1994. Introduction to Agrometeorology. Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi.
6. H.V. Nanjappa and B.K. Ramachandrappa. 2007. Manual on Practical Agricultural Meteorology. Agrobios India. Jodhpur.
7. S.R. Reddy. 1999. Principles of Agronomy. Kalyani Publishers, New Delhi.
8. T. Yellamanda Reddy and G.H. Sankara Reddi. 2010. Principles of Agronomy. Kalyani Publishers, New Delhi.

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|------------------|-----------------------------------|---------------|
| <b>AGRON-322</b> | <b>Introductory Agro-forestry</b> | <b>2(1+1)</b> |
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#### **Objectives**

1. To study Agro forestry as an alternate system of land use
2. To study different types of Agro forestry for soil and water conservation.
3. To study the characteristics of Agro forestry in terms its potential for soil moisture conservation Practices

#### **Theory**

Agro-forestry: Definition and scope of Agroforestry system, Type of Agroforestry system, potential of Agroforestry in India, Prevailing agroforestry system in India; MPTS- definition,

role of MPTS in agroforestry system, its selection for different agroforestry system, MPTS of India, Ecological aspects of Agroforestry system, tree -crop interaction – competition, nutrient recycling; Traditional Agroforestry as a viable choice to conserve Agro biodiversity of India. Management of Agro-forestry system; Role of agroforestry in soil and water conservation; windbreak; Shelterbelt– definition, objectives.; Socio- economic aspects of Agroforestry system; Design and Diagnostic study of agroforetry system; Silviculture: Definition and scope, Propagation of tree species, Regeneration by seed, coppice, root suckers, Transplanting, stump, branch cutting, rhizomes; Nursery bed preparation and management; Cultural practices for bare root and seedling, field handling of nursery stock; Management of tree species; Silviculture of important tree species, choice of species- site factors, root, crown and bole characteristics, phenology, nutritional and water requirement, ground operation, tending, harvesting utility etc. Horticulture and forage crops-based agroforestry models developed by ICAR-IGFRI; Agroforestry models developed by Indian council of Forestry Research and Education.

### **Practical**

Identification of tree species in agro-forestry, Study of tree growth measurement, Study of environmental parameters affecting AF System, Plant propagation methods, Pre-sowing seed treatment, Preparation of nursery bed exercise, practicing propagation techniques for trees, Afforestation method, practical training, pruning, coppicing, pollarding etc. Planting pattern and designs for plantation, natural and artificial regeneration, Design and diagnostic survey of agroforestry system, Evaluation of agro-forestry system in different agro climatic zones, Exposure Visit to prevailing agroforestry systems of the state and related important institutions, Virtual visit of agroforestry models developed by ICAR-IGFRI, ICFRE.

### **Lecture schedule: Theory**

| S.N. | Topic                                                                                                                 | No. of lectures |
|------|-----------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.   | Agro-forestry: Definition and scope of Agroforestry system, Type of Agroforestry system                               | 1               |
| 2.   | Potential of Agroforestry in India, Prevailing agroforestry system in India                                           | 1               |
| 3.   | MPTS- definition, role of MPTS in agroforestry system, its selection for different agroforestry system, MPTS of India | 1               |
| 4.   | Ecological aspects of Agroforestry system; Tree -crop interaction – competition, nutrient recycling                   | 1               |
| 5.   | Traditional Agroforestry as a viable choice to conserve Agro biodiversity of India.                                   | 1               |

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| 6.  | Management of Agro-forestry system; Role of agroforestry in soil and water conservation; Windbreak; Shelterbelt– definition, objectives.                                | 1 |
| 7.  | Socio- economic aspects of Agroforestry system; Design and Diagnostic study of agroforestry system                                                                      | 1 |
| 8.  | Silviculture: Definition and scope; Propagation of tree species, Regeneration by seed, coppice, root suckers, Transplanting, stump, branch cutting, rhizomes;           | 1 |
| 9.  | Nursery bed preparation and management                                                                                                                                  | 1 |
| 10. | Cultural practices for bare root and seedling, field handling of nursery stock                                                                                          | 1 |
| 11. | Management of tree species                                                                                                                                              | 1 |
| 12. | Silviculture of important tree species                                                                                                                                  | 1 |
| 13. | Choice of species- site factors, root, crown and bole characteristics, phenology, nutritional and water requirement, ground operation, tending, harvesting utility etc. | 2 |
| 14. | Horticulture and forage crops-based agroforestry models developed by ICAR-IGFRI;                                                                                        | 1 |
| 15. | Agroforestry models developed by Indian council of Forestry Research and Education.                                                                                     | 1 |

**Lecture schedule: Practical**

| S.N. | Topic                                                                                             | No. of lectures |
|------|---------------------------------------------------------------------------------------------------|-----------------|
| 1.   | Identification of tree species in agro-forestry                                                   | 1               |
| 2.   | Study of tree growth measurement                                                                  | 1               |
| 3.   | Study of environmental parameters affecting AF System                                             | 1               |
| 4.   | Plant propagation methods, Pre-sowing seed treatment                                              | 1               |
| 5.   | Preparation of nursery bed exercise                                                               | 1               |
| 6.   | practicing propagation techniques for trees                                                       | 1               |
| 7.   | Afforestation method, practical training, pruning, coppicing, pollarding etc.                     | 2               |
| 8.   | Planting pattern and designs for plantation, natural and artificial regeneration                  | 2               |
| 9.   | Design and diagnostic survey of agroforestry system                                               | 1               |
| 10.  | Evaluation of agro-forestry system in different agro climatic zones                               | 1               |
| 11.  | Exposure Visit to prevailing agroforestry systems of the state and related important institutions | 2               |
| 12.  | Virtual visit of agroforestry models developed by ICAR-IGFRI, ICFRE.                              | 2               |

**Suggested readings**

1. Nair, P.K. R. 1993. An Introduction to Agroforestry, Kluar Academic Publisher
2. Chundawat D. S. and S.K. Gautham. 2017. Textbook of Agroforestry. Oxford & IBH Publishing, (ISBN: 9788120408326)
3. Parthiban, K. T, N. Krishnakumar and M. Karthick. 2018. Introduction to Forestry, Scientific Publisher, Jodhpur. 350p
4. Divya M. P. and K. T. Parthiban. 2005. A Textbook on Social Forestry and Agroforestry. Satish Serial Publishing, New Delhi (ISBN: 9384988952).

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|------------------|----------------------------------------------|---------------|
| <b>AECON-321</b> | <b>Agricultural Finance and Co-Operation</b> | <b>2(1+1)</b> |
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**Course Objective**

1. To impart knowledge on issues related to lending to priority sector credit management and financial risk management.
2. To impart knowledge about crop insurance and its role in minimizing farm risk.

**Theory**

Agricultural Finance-meaning, scope and significance, credit needs and its role in Indian agriculture. Agricultural credit: meaning, definition, need, classification. Credit analysis: 3 R's, 5C's and 7 P's of credits. Sources of agricultural finance: institutional and non- institutional sources, commercial banks, social control and nationalization of commercial banks. Micro financing including KCC. Lead bank scheme, RRBs, Scale of finance and unit cost. An introduction to higher financing institutions – RBI, NABARD, ADB, IMF, world bank, Insurance and Credit Guarantee Corporation of India. Cost of credit. Recent development in agricultural credit. Preparation and analysis of financial statements – Balance Sheet and Income Statement. Basic guidelines for preparation of project reports. Bank norms – SWOT analysis. Agricultural Cooperation – Meaning, brief history of cooperative development in India, objectives, principles of cooperation, significance of cooperatives in Indian agriculture. Agricultural Cooperation in India- credit, marketing, consumer and multi-purpose cooperatives, farmers' service cooperative societies, processing cooperatives, farming cooperatives, cooperative warehousing; role of ICA, NCUI, NCDC, NAFED. Crop insurance: its scope, significance and limitations and the potential of the newly launched 'Pradhan Mantri Fasal Bima Yojana' (Prime Minister's Crop Insurance Scheme). Successful cooperative systems in Gujarat (AMUL), Tamil Nadu (Aavin), Karnataka (Nandini), Maharashtra and Punjab.

**Practical**

Optimum allocation of limited amount of capital among different enterprises. Analysis of progress and performance of cooperatives using published data. Analysis of progress and performance of commercial banks and RRBs using published data. Visit to a commercial bank, cooperative bank and cooperative society to acquire first-hand knowledge of their management, schemes and procedures. Estimation of credit requirement of farm business – A case study.

Preparation and analysis of balance sheet – A case study. Preparation and analysis of income statement–A case study. Appraisal of a loan proposal –A case study. Techno-economic parameters for preparation of projects. Preparation of Bankable projects for various agricultural products and its value-added products. Seminar on selected topics. Different types of repayment plans.

### Lecture Schedule: Theory

| S. No. | Name of Topic                                                                                                                                                                    | No. of Lectures |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.     | Agricultural Finance-meaning, scope and significance, credit needs and its role in Indian agriculture.                                                                           | 1               |
| 2.     | Agricultural credit: meaning, definition, need, classification. Credit analysis: 3 R's, 5C's and 7 P's of credits.                                                               | 2               |
| 3.     | Sources of agricultural finance: institutional and non- institutional sources                                                                                                    | 1               |
| 4.     | Commercial banks, social control and nationalization of commercial banks. Micro financing including KCC.                                                                         | 1               |
| 5.     | Lead bank scheme, RRBs, Scale of finance and unit cost.                                                                                                                          | 1               |
| 6.     | An introduction to higher financing institutions – RBI, NABARD, ADB, IMF                                                                                                         | 1               |
| 7.     | World bank, Insurance and Credit Guarantee Corporation of India. Cost of credit, Recent development in agricultural credit                                                       | 1               |
| 8.     | Preparation and analysis of financial statements – Balance Sheet and Income Statement.                                                                                           | 1               |
| 9.     | Basic guidelines for preparation of project reports. Bank norms – SWOT analysis.                                                                                                 | 1               |
| 10.    | Agricultural Cooperation – Meaning, brief history of cooperative development in India, Objectives, principles of cooperation, significance of cooperatives in Indian agriculture | 1               |
| 11.    | Agricultural Cooperation in India- credit, marketing, consumer and multi-purpose cooperatives                                                                                    | 1               |
| 12.    | Farmers' service cooperative societies, processing cooperatives, farming cooperatives                                                                                            | 1               |
| 13.    | Cooperative warehousing; role of ICA, NCUI, NCDC, NAFED                                                                                                                          | 1               |
| 14.    | Crop insurance: its scope, significance and limitations and the potential of the newly launched 'Pradhan Mantri Fasal Bima Yojana' (Prime Minister's Crop Insurance Scheme).     | 1               |



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| 15. | Successful cooperative systems in Gujarat (AMUL), Tamil Nadu (Aavin), Karnataka (Nandini), Maharashtra and Punjab and Rajasthan | 1 |
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**Lecture Schedule: Practical**

| S. No. | Name of Topic                                                                                                                                     | No. of Lectures |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.     | Optimum allocation of limited amount of capital among different enterprise.                                                                       | 1               |
| 2.     | Analysis of progress and performance of cooperatives using published data.                                                                        | 1               |
| 3.     | Analysis of progress and performance of commercial banks and RRBs using published data.                                                           | 1               |
| 4.     | Visit to a commercial bank, cooperative bank and cooperative society to acquire first-hand knowledge of their management, schemes and procedures. | 2               |
| 5.     | Estimation of credit requirement of farm business – A case study.                                                                                 | 1               |
| 6.     | Preparation and analysis of balance sheet – A case study.                                                                                         | 1               |
| 7.     | Preparation and analysis of income statement – A case study.                                                                                      | 1               |
| 8.     | Appraisal of a loan proposal – A case study.                                                                                                      | 2               |
| 9.     | Techno-economic parameters for preparation of projects.                                                                                           | 2               |
| 10.    | Preparation of Bankable projects for various agricultural products and its value-added products                                                   | 1               |
| 11.    | Seminar on selected topics                                                                                                                        | 2               |
| 12.    | Different types of repayment plans.                                                                                                               | 1               |

**Suggested readings**

1. Gittinger, J. P. 1982. Economic Analysis of Agricultural Projects. The Johns Hopkins Univ. Press.
2. Reddy, S. S. and Ram, P. R. 1996. Agricultural Finance and Management. Oxford & IBH.

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| <b>BIOCH-321</b> | <b>Essentials of Plant Biochemistry</b> | <b>3(2+1)</b> |
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**Theory:**

Biochemistry – Introduction and importance, Properties of water, pH and buffer, plant cell and its components. Bio-molecules – Structure, classification, properties and function of carbohydrates, amino acids, proteins, lipids and nucleic acids. Vitamins – physiological and metabolic role. Enzymes: General properties; Classification; Mechanism of action; Michaelis and Menten and Line Weaver Burk equation and plots; Introduction to allosteric enzymes, use of

enzymes. Metabolic energy and its generation – Metabolism – Basic concepts, Glycolysis, Citric acid Cycle, Pentose phosphate pathway, oxidative phosphorylation, Fatty acid oxidation. Biosynthetic Pathways – Photosynthesis, Gluconeogenesis, nitrogen fixation, fatty acid and starch formation. Regulation of metabolic pathways. Secondary metabolites, Terpenoids, Alkaloids, Phenolic and their applications in food and pharmaceutical industries.

### Practical:

Preparation of standard solutions and reagents, Determination of pH, Qualitative tests of carbohydrates and amino acids, Quantitative estimation of soluble sugars and starch, Estimation of protein by Kjeldhal method and Lowry's method, Preparation of mineral solution from ash, Estimation of fat by Soxhlet method, Determination of acid value, saponification value and iodine number, Estimation of ascorbic acid, Qualitative/quantitative tests of secondary

### Theory Schedule

| Class | Topic                                                                         |
|-------|-------------------------------------------------------------------------------|
| 1     | Introduction to Biochemistry and its Importance                               |
| 2     | Properties of Water, pH, and Buffer Systems                                   |
| 3     | Plant Cell and Its Components                                                 |
| 4     | Carbohydrates: Structure, Classification, Properties, and Function            |
| 5     | Amino Acids: Structure, Classification, Properties, and Function              |
| 6     | Proteins: Structure, Classification, Properties, and Function                 |
| 7     | Lipids: Structure, Classification, Properties, and Function                   |
| 8     | Nucleic Acids: Structure, Classification, Properties, and Function            |
| 9     | Vitamins: Physiological and Metabolic Role                                    |
| 10    | Enzymes: General Properties and Classification                                |
| 11    | Enzyme Mechanism of Action                                                    |
| 12    | Michaelis-Menten Equation and Lineweaver-Burk Plot                            |
| 13    | Allosteric Enzymes and Their Functions                                        |
| 14    | Use of Enzymes in Biotechnology and Industry                                  |
| 15    | Metabolic Energy: Basic Concepts                                              |
| 16    | Glycolysis and Its Role in Energy Generation                                  |
| 17    | Citric Acid Cycle (TCA Cycle)                                                 |
| 18    | Pentose Phosphate Pathway                                                     |
| 19    | Oxidative Phosphorylation and ATP Generation                                  |
| 20    | Fatty Acid Oxidation                                                          |
| 21    | Photosynthesis: Light Reactions and Calvin Cycle                              |
| 22    | Gluconeogenesis and Its Regulation                                            |
| 23    | Nitrogen Fixation: Mechanism and Significance                                 |
| 24    | Fatty Acid Biosynthesis and Starch Formation                                  |
| 25    | Regulation of Metabolic Pathways: Primary and secondary metabolites of plants |

| Class | Topic                                                            |
|-------|------------------------------------------------------------------|
| 26    | Secondary Metabolites: Terpenoids – Structure and Function       |
| 27    | Secondary Metabolites: Alkaloids – Structure and Function        |
| 28    | Secondary Metabolites: Phenolics – Structure and Function        |
| 29    | Applications of Terpenoids in Food and Pharmaceutical Industries |
| 30    | Applications of Alkaloids in Food and Pharmaceutical Industries  |
| 31    | Applications of Phenolics in Food and Pharmaceutical Industries  |
| 32    | Review of Biochemistry and Summary                               |

### Practical Schedule

| Class | Topic                                                                        |
|-------|------------------------------------------------------------------------------|
| 1     | Preparation of Standard Solutions and Reagents                               |
| 2     | Determination of pH                                                          |
| 3     | Qualitative Tests for Carbohydrates                                          |
| 4     | Qualitative Tests for Amino Acids                                            |
| 5     | Quantitative Estimation of TSS                                               |
| 6     | Quantitative Estimation of Starch                                            |
| 7     | Estimation of Protein by Kjeldhal Method                                     |
| 8     | Estimation of Protein by Lowry's Method                                      |
| 9     | Preparation of Mineral Solution from Ash                                     |
| 10    | Estimation of Fat by Soxhlet Method                                          |
| 11    | Determination of Acid Value of Lipids                                        |
| 12    | Determination of Saponification Value                                        |
| 13    | Determination of Iodine Number of Lipids                                     |
| 14    | Estimation of Ascorbic Acid                                                  |
| 15    | Qualitative/ Quantitative Tests for Secondary Metabolites                    |
| 16    | Estimation of Post-harvest and stress-mediated biochemical changes in fruits |

### Suggested reading:

- David L. Nelson, Michael Cox. 2021. *Lehninger Principles of Biochemistry*. Eighth edition. W. H. Freeman & Co Ltd
- Conn, Stumpf, Bruening and Doi. 2006. *Outlines of Biochemistry*. Fifth Edition. Wiley
- Horton, Moran, Rawn, Scrimgeour, Perry. 2011. *Principles of Biochemistry*. Fifth Edition. Pearson/Prentice Hall
- Heldt. 2005. *Plant Biochemistry*. Elsevier (Can be downloaded)
- Goodwin and Mercer. 2005. *Introduction to Plant Biochemistry*. 2nd edition. CBS.
- U Satyanarayana. 2013. *Biochemistry* 4th Edition. Elsevier India

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| BT-321 | Fundamentals of Agriculture Biotechnology | 3(2+1) |
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**Theory:**

Introduction to Plant Tissue Culture and Genetic Engineering: History; Cellular totipotency and cytodifferentiation; Composition of tissue culture media; Callus culture, Single-cell/suspension culture and their applications; Organogenesis and somatic embryogenesis; Somaclonal variation and its use in crop improvement; Embryo rescue technique and its significance in hybrid development; *In vitro* fertilization, ovule culture and its significance in hybrid development; Role of PGRs in tissue culture; Protoplast isolation, culture and regeneration; Somatic hybridization (somatic hybrids and cybrids) and its application in crop improvement; Anther and pollen culture for haploid production; Development of disease-free (virus free) plants through apical meristem culture; Micropropagation technique for the generation of quality planting material; Synthetic seeds and its applications; National certification and Quality management of TC plants-secondary metabolite production- *in vitro* germplasm conservation.

Introduction to Molecular Biology: DNA structure and function; DNA replication, transcription and translation; RNA, types and function; Structure of prokaryotic and eukaryotic gene; Central dogma of life - DNA replication, transcription, genetic codes-translation and protein synthesis; Lac Operon concept - Nucleic acid hybridization; Polymerase chain reaction- DNA sequencing – Sanger method; PCR and its applications.

Introduction to recombinant DNA technology: DNA modifying enzymes and vectors; plant genetic transformation – physical (Gene gun method), chemical (PEG mediated) and *Agrobacterium*-mediated gene transfer methods; Transgenic and its importance in crop improvement with successful stories; biosafety. Introduction to various molecular markers: RFLP, RAPD, SSR, SNP etc.; Marker-assisted breeding in crop improvement

**Practical:**

Introduction to Plant Tissue Culture Laboratory; Good Laboratory Practices; Media Preparation and sterilization; Glassware sterilization; Micropropagation; Callus induction and culture; *In vitro* morphogenesis; Apical meristem culture; Preparation of synthetic seeds; Isolation of plasmid DNA; Quantification of DNA; Agarose Gel Electrophoresis and visualization of plasmid DNA; Restriction digestion of plasmid DNA and agarose gel electrophoresis; Isolation of Plant genomic DNA; PCR amplification of DNA; Gel electrophoresis of amplified DNA; Visit to tissue culture units /biotech labs.

**Theory Schedule**

| Class | Topic |
|-------|-------|
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| Class | Topic                                                                  |
|-------|------------------------------------------------------------------------|
| 1     | Introduction to Plant Tissue Culture and Genetic Engineering: History  |
| 2     | Cellular Totipotency and Cytodifferentiation                           |
| 3     | Composition of tissue culture media                                    |
| 4     | Callus Culture, Single Cell/Suspension Culture, and their applications |
| 5     | Organogenesis and Somatic Embryogenesis                                |
| 6     | Somaclonal Variation and its use in Crop Improvement                   |
| 7     | Embryo Rescue Technique and Hybrid Development                         |
| 8     | <i>In Vitro</i> Fertilization and Ovule Culture                        |
| 9     | Role of PGRs in tissue culture                                         |
| 10    | Protoplast Isolation, Culture, and Regeneration                        |
| 11    | Somatic Hybridization (Somatic Hybrids and Cybrids)                    |
| 12    | Anther and Pollen Culture for Haploid Production                       |
| 13    | Development of Disease-Free Plants through Apical Meristem Culture     |
| 14    | Micropropagation for Quality Planting Material                         |
| 15    | Synthetic Seeds and its Applications                                   |
| 16    | National Certification and Quality Management of Tissue Culture Plants |
| 17    | Secondary Metabolite Production                                        |
| 18    | <i>In Vitro</i> Germplasm Conservation                                 |
| 19    | Introduction to Molecular Biology: DNA Structure and Function          |
| 20    | RNA: Types and Functions                                               |
| 21    | Structure of Prokaryotic and Eukaryotic Genes                          |
| 22    | Central Dogma of life: DNA Replication, Transcription, and Translation |
| 23    | Lac Operon Concept and Genetic Codes                                   |
| 24    | Polymerase Chain Reaction (PCR) and its applications                   |
| 25    | Introduction to Recombinant DNA Technology: Enzymes and Vectors        |
| 26    | Plant Genetic Transformation: Physical and Chemical Methods            |
| 27    | <i>Agrobacterium</i> -Mediated Gene Transfer                           |
| 28    | Transgenic Plants and Their Importance in Crop Improvement             |
| 29    | Biosafety and Ethical Considerations                                   |
| 30    | Introduction to Molecular Markers: RFLP, RAPD, SSR, SNP                |
| 31    | Marker-Assisted Breeding in Crop Improvement                           |
| 32    | Course Review and summary                                              |

### Practical Schedule

| Class | Topic                                                         |
|-------|---------------------------------------------------------------|
| 1     | Introduction to Plant Tissue Culture Laboratory and equipment |
| 2     | Media Preparation and Sterilization                           |

| Class | Topic                                         |
|-------|-----------------------------------------------|
| 3     | Glassware Sterilization                       |
| 4     | Micropropagation Techniques                   |
| 5     | Callus Induction and Culture                  |
| 6     | <i>In vitro</i> morphogenesis                 |
| 7     | Apical Meristem Culture                       |
| 8     | Preparation of Synthetic Seeds                |
| 9     | Isolation of Plasmid DNA                      |
| 10    | Quantification of DNA                         |
| 11    | Agarose Gel Electrophoresis and Visualization |
| 12    | Restriction Digestion of Plasmid DNA          |
| 13    | Isolation of Plant Genomic DNA                |
| 14    | PCR Amplification of DNA                      |
| 15    | Gel Electrophoresis of Amplified DNA          |
| 16    | Visit to Tissue Culture Units/Biotech Labs    |

**Suggested readings:**

1. Bhojwani SS. 1983. Plant Tissue Culture: Theory and Practice. Elsevier.
2. Singh BD. 2007. Biotechnology: Expanding Horiozon. Kalyani
3. Christou P and Klee H. 2004. Handbook of Plant Biotechnology. John Wiley & Sons.
4. Lewin B. 2008. Gene IX. Peterson Publications/ Panima. W.H. Freeman & Co.
5. Primrose SB. 2001. Molecular Biotechnology. Panima.
6. Chawla H.S. Third edition. Introduction to Plant Biotechnology. Oxford & IBH, New Delhi

|                |                                                |               |
|----------------|------------------------------------------------|---------------|
| <b>GPB-321</b> | <b>Crop Improvement (<i>Rabi</i> crops)-II</b> | <b>2(1+1)</b> |
|----------------|------------------------------------------------|---------------|

**Objectives:**

1. To provide knowledge about self-pollinated and cross-pollinated *rabi* crops
2. To learn about origin and distribution of *rabi* crops
3. To design breeding objectives of major *rabi* crops
4. To impart information on different crop varieties for *rabi* season

**Theory**

Centres of origin, distribution of species, wild relatives in different cereals; pulses; oilseeds; fibres; fodders and cash crops; vegetable and other horticultural crops; Plant genetic resources, its utilization and conservation, study of genetics of qualitative and quantitative characters; Important concepts of breeding self-pollinated, cross-pollinated and vegetatively propagated

crops. Major breeding objectives and procedures including conventional and modern innovative approaches for development of hybrids and varieties for yield, adaptability, stability, abiotic and biotic stress tolerance and quality (physical, chemical, nutritional); Hybrid seed production technology in wheat, oat, chickpea, rapeseed and mustard etc. Ideotype concept, climate resilient crop varieties for future.

### **Practical**

Botany of crops, Floral biology, emasculation and hybridization techniques in different crop species, viz. wheat, oat, rapeseed and mustard, pulses, potato, sugarcane, tomato, chilli, onion etc. Study of field techniques for seed production and hybrid seed production in rabi crops; Estimation of heterosis, inbreeding depression and heritability; Study of quality characters, donor parents for different characters; Visit to seed production plots; Visit to AICRP breeding plots of different crops.

### **Lecture Schedule: Theory**

| S.N. | Topic                                                                                                                                                                        | No. of lectures |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1    | Crop improvement aspects in wheat as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.     | 1               |
| 2    | Crop improvement aspects in oat as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.       | 1               |
| 3    | Crop improvement aspects in barley as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.    | 1               |
| 4    | Crop improvement aspects in chickpea as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.  | 1               |
| 5    | Crop improvement aspects in lentil as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.    | 1               |
| 6    | Crop improvement aspects in field pea as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc. | 1               |

|    |                                                                                                                                                                                                             |   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 7  | Crop improvement aspects in rapeseed mustard as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.                         | 1 |
| 8  | Crop improvement aspects in rapeseed mustard as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc & hybrid seed production | 1 |
| 9  | Crop improvement aspects in sunflower as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.                                | 1 |
| 10 | Crop improvement aspects in berseem as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc & hybrid seed production          | 1 |
| 11 | Crop improvement aspects in lucerne as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc & hybrid seed production          | 1 |
| 12 | Crop improvement aspects in sugarcane as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.                                | 1 |
| 13 | Modern innovative approaches for development of hybrids and varieties for yield, adaptability, stability, abiotic and biotic stress tolerance and quality (physical, chemical, nutritional)                 | 1 |
| 14 | Seed production technology in self pollinated, cross pollinated and vegetatively propagated crops                                                                                                           | 1 |
| 15 | Ideotype concept                                                                                                                                                                                            | 1 |
| 16 | Climate resilient crop varieties for future                                                                                                                                                                 | 1 |

### **Lecture Schedule: Practical**

| <b>S.N.</b> | <b>Topic</b>                                                             | <b>No. of lectures</b> |
|-------------|--------------------------------------------------------------------------|------------------------|
| 1           | Emasculation and hybridization techniques in wheat, oats, barley         | 1                      |
| 2           | Emasculation and hybridization techniques in chickpea, lentil, field pea | 1                      |
| 3           | Emasculation and hybridization techniques in rapeseed mustard            | 1                      |
| 4           | Emasculation and hybridization techniques in sunflower, potato           | 1                      |
| 5           | Emasculation and hybridization techniques in berseem, sugarcane          | 1                      |
| 6           | Maintenance breeding of different rabi crops                             | 1                      |



|    |                                                                                                                           |   |
|----|---------------------------------------------------------------------------------------------------------------------------|---|
| 7  | Handling of germplasm and segregating populations by different methods like pedigree, bulk and single seed decent methods | 1 |
| 8  | Study of field techniques for seed production and hybrid seeds production in <i>rabi</i> crops                            | 1 |
| 9  | Estimation of heterosis, inbreeding depression and heritability                                                           | 1 |
| 10 | Layout of field experiments                                                                                               | 1 |
| 11 | Study of quality characters                                                                                               | 1 |
| 12 | Donor parents for different characters                                                                                    | 1 |
| 13 | Visit to seed production plots                                                                                            | 2 |
| 14 | Visit to AICRP plots of different field crops                                                                             | 2 |

### Suggested readings

1. Breeding Field Crops -I by V.L. Chopra
2. Genetic Improvement of Field Crops by C.B. Singh and D. Khare
3. Genetics and Breeding of Pulse crops by D.P. Singh
4. Vegetable Breeding – Principles and Practices by Hari Har Ram
5. Breeding Field Crops by D.A. Sleper and J.M. Poehlman
6. Plant Breeding –Theory and practice by S.K. Gupta
7. Breeding Asian field Crops by J.M. Poehlman and D.N. Barthakur
8. Practical Manuals on Crop Improvement I (*Rabi* crops) by Rajendra Kumar Yadav

|                |                                                    |               |
|----------------|----------------------------------------------------|---------------|
| <b>GPB-322</b> | <b>Fundamentals of Seed Science and Technology</b> | <b>2(1+1)</b> |
|----------------|----------------------------------------------------|---------------|

### Objectives

1. To impart basic and fundamental knowledge on principles and practices seed science and technology
2. To impart practical skills on scientific seed production and post-harvest quality management

### Theory

Introduction to seed technology, definition and importance; Seed quality -definition, characters of good quality seed; Causes of deterioration of varietal purity and assessment of genetic purity, different classes of seed. Foundation and certified seed production of important cereals, pulses and oilseed, field inspection, importance and procedures; Post-harvest seed quality management; seed processing procedures, seed drying; Seed treatment, its importance, method of application and seed packing; seed storage - general principles, stages and factors affecting seed longevity during storage; Seed health management during storage. Seed Certification and legislation; Seed Act and Seed Act enforcement, duty and powers of seed inspector, offences and penalties. Seeds Control Order 1983, basics of seed quality testing; New Seed Bill 2019; Seed quality enhancement techniques.

**Practical**

Seed Structure, Seed sampling, Physical purity, Moisture determination, Germination test, Seed and seedling vigour test, Seed Viability, Genetic purity test: Grow out test, Field inspection, Seed health testing using blotter and agar plate method. Visit to seed production farms, seed testing laboratories and seed processing plant.

**Lecture Schedule: Theory**

| S.N<br>. | Topic                                                                                                                                                | No. of<br>lectures |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 1        | Seed and seed technology: introduction, definition and importance                                                                                    | 1                  |
| 2        | Deterioration causes of crop varieties and their control; Maintenance of genetic purity during seed production                                       | 1                  |
| 3        | Seed quality; Definition, Characters of good quality seed, different classes of seed                                                                 | 1                  |
| 4        | Foundation and certified seed production of important cereals & fodder                                                                               | 1                  |
| 5        | Foundation and certified seed production of important pulses                                                                                         | 1                  |
| 6        | Foundation and certified seed production of important oilseeds                                                                                       | 1                  |
| 7        | Foundation and certified seed production of important vegetables                                                                                     | 1                  |
| 8        | Foundation and certified seed production of important seed spices                                                                                    | 1                  |
| 9        | Seed certification, phases of certification, procedure for seed certification, field inspection                                                      | 1                  |
| 10       | Seed Act and Seed Act enforcement. Duty and powers of seed inspector, offences and penalties. Seeds Control Order 1983                               | 1                  |
| 11       | Varietal Identification through Grow Out Test and Electrophoresis, Molecular and Biochemical test                                                    | 1                  |
| 12       | Detection of genetically modified crops, Transgene contamination in non-GM crops                                                                     | 1                  |
| 13       | GM crops and organic seed production                                                                                                                 | 1                  |
| 14       | Seed drying, processing and their steps, seed testing for quality assessment, seed treatment, its importance, method of application and seed packing | 1                  |
| 15       | Seed storage; general principles, stages and factors affecting seed longevity during storage. Measures for pest and disease control during storage   | 1                  |

|    |                                                                                                                                                                      |   |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 16 | Seed marketing: structure and organization, sales generation activities, promotional media. Factors affecting seed marketing, Role of WTO and OECD in seed marketing | 1 |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

### Lecture Schedule: Practical

| S.N<br>. | Topic                                                                                                                                                                     | No. of<br>lectures |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 1        | Seed production in wheat including seed standards                                                                                                                         | 1                  |
| 2        | Seed production in rice including seed standards                                                                                                                          | 1                  |
| 3        | Seed production in Maize including seed standards                                                                                                                         | 1                  |
| 4        | Seed production in Sorghum including seed standards                                                                                                                       | 1                  |
| 5        | Seed production in Bajra including seed standards                                                                                                                         | 1                  |
| 6        | Seed production in Urd, Mung and Cowpea including seed standards                                                                                                          | 1                  |
| 7        | Seed production in Pigeonpea including seed standards                                                                                                                     | 1                  |
| 8        | Seed production in Lentil including seed standards                                                                                                                        | 1                  |
| 9        | Seed production in Gram including seed standards                                                                                                                          | 1                  |
| 10       | Seed production in Field pea including seed standards                                                                                                                     | 1                  |
| 11       | Seed production in Soybean including seed standards                                                                                                                       | 1                  |
| 12       | Seed production in Rapeseed and Mustard including seed standards                                                                                                          | 1                  |
| 13       | Seed production in Groundnut and Sesame including seed standards                                                                                                          | 1                  |
| 14       | Seed production in vegetable crops (Potato, cauliflower, tomato and chilli) including seed standards                                                                      | 1                  |
| 15       | Seed production in Seed spices (fenugreek, fennel, cumin & coriander) including seed standards                                                                            | 1                  |
| 16       | Visit to seed testing laboratories , Visit to seed production farms, seed processing plant, Seed sampling methods, Physical purity test, Germination test, Viability test | 1                  |

### Suggested Readings

1. Agarwal, R.L. 1995. Seed Technology (2<sup>nd</sup> edition). Oxford & IBH Publishing Co. Pvt. Ltd. New Delhi, India.
2. Khare, D. and Bhale, M.S. 2019. Seed Technology (2<sup>nd</sup> revised & enlarged edn), Scientific Publishers, ISBN: 978-81-72338-84-8, New Pali Road, P.O. Box 91, Jodhpur, India
3. Vanangamudi, K. 2014. Seed Technology (An illustrated book), New India Publishing Agency, New Delhi, India.
4. Bhojwani, S.S. and Bhatnagar, S.P. 1999. The Embryology of Angiosperm. Vikas Publ

5. McDonald, M.B. Jr and Copeland, L.O. 1997. Seed Production: Principles and Practices. Chapman & Hall.

|                  |                                                        |               |
|------------------|--------------------------------------------------------|---------------|
| <b>PPATH-321</b> | <b>Agricultural Microbiology and Phyto-remediation</b> | <b>2(1+1)</b> |
|------------------|--------------------------------------------------------|---------------|

### Objectives

1. To get an introduction to microbiology with specific focus on its significance in agriculture science
2. To get acquainted with the bacterial structure and the function of the different bacterial components
3. To get highlights on different fields of microbiology
4. To get highlights on the bioremediation of polluted soils using microbial mediators and Phytoremediation
5. To get a concept of biological control and the role of biopesticides in plant disease management.

### Theory

Introduction to Microbiology: Definition, applied areas of Microbiology and Importance of Microbiology. History of Microbiology: Discovery of microorganisms, spontaneous generation theory, Germ theory of diseases, Immunization, fermentation, and origin of life. Bacteria: cell structure, chemoautotrophy, photo autotrophy, growth. Bacterial genetics: Genetic recombination- transformation, conjugation and transduction, genetic engineering. Soil Microbiology: Nutrient mineralization and transformation, Air Microbiology: Phyllo sphere microflora, Phylloplane microflora, microflora of floral parts etc. Food Microbiology: Microbial spoilage and principles of food preservations, Food poisoning. Water Microbiology: Types of water, water microorganisms, and microbial analysis of water e.g. coliform test, Purification of water. Industrial Microbiology: Microbial products, Biodegradation, Biogas production, Biodegradable plastics etc. Biological control: Microbial biopesticides for plant disease management Concepts of rhizosphere microbiology- Rhizospheres -biochemical nature, release mechanism in rhizosphere, function, Carbon flow in rhizosphere, Rhizosphere microbiome residents and their roles. Potential of plant growth promoting rhizobacteria (PGPR) and endophytes on soil health and sustainability. Bioremediation of polluted soils using microbial mediators. Phytoremediation of polluted soils.

### Practical

Study of the microscope; Acquaintance with laboratory material and equipment; Microscopic observation of different groups of microorganisms: moulds (Fungi); Direct staining of bacteria by crystal violet; Negative or indirect staining of bacteria by nigrosin; Gram staining of bacteria; Study of phyllosphere and rhizosphere microflora; Measurement of microorganisms; Preparation

of culture media; Isolation and purification of rhizospheric microbes; Isolation and purification of N-fixers; Isolation and purification of Nutrient solubilizers; Isolation and purification of Endophytes.

### Lecture Schedule: Theory

| S.No. | Topic                                                                                                                                                                                              | No. of lectures |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.    | Introduction to Microbiology: Definition, applied areas of Microbiology and Importance of Microbiology.                                                                                            | 01`             |
| 2.    | History of Microbiology: Discovery of microorganisms, spontaneous generation theory, Germ theory of diseases, Immunization, fermentation, and origin of life.                                      | 01              |
| 3.    | Bacteria: cell structure, chemoautotrophy, photo autotrophy, growth.                                                                                                                               | 01              |
| 4.    | Bacterial genetics: Genetic recombination- transformation, conjugation and transduction, genetic engineering.                                                                                      | 02              |
| 5.    | Soil Microbiology: Nutrient mineralization and transformation, Air Microbiology: Phyllosphere microflora, Phylloplane microflora, microflora of floral parts etc.                                  | 02              |
| 6.    | Food Microbiology: Microbial spoilage and principles of food preservations, Food poisoning.                                                                                                        | 01              |
| 7.    | Water Microbiology: Types of water, water microorganisms, and microbial analysis of water e.g. coli form test, Purification of water.                                                              | 02              |
| 8.    | Industrial Microbiology: Microbial products, Biodegradation, Biogas production, Biodegradable plastics etc.                                                                                        | 01              |
| 9.    | Biological control: Microbial biopesticides for plant disease management.                                                                                                                          | 01              |
| 10.   | Concepts of rhizosphere microbiology- Rhizodeposits -biochemical nature, release mechanism in rhizosphere, function, Carbon flow in rhizosphere, Rhizosphere microbiome residents and their roles. | 02              |
| 11.   | Potential of plant growth promoting rhizobacteria (PGPR) and endophytes on soil health and sustainability.                                                                                         | 01              |
| 12.   | Bioremediation of polluted soils using microbial mediators. Phytoremediation of polluted soils.                                                                                                    | 01              |

### Lecture Schedule: Practical

| S.No. | Topic                                               | No. of lectures |
|-------|-----------------------------------------------------|-----------------|
| 1.    | Study of the microscope                             | 02              |
| 2.    | Acquaintance with laboratory material and equipment | 02              |

|     |                                                                                             |    |
|-----|---------------------------------------------------------------------------------------------|----|
| 3.  | Microscopic observation of different groups of microorganisms: moulds (Fungi)               | 01 |
| 4.  | Direct staining of bacteria by crystal violet                                               | 01 |
| 5.  | Negative or indirect staining of bacteria by nigrosin                                       | 01 |
| 6.  | Gram staining of bacteria                                                                   | 01 |
| 7.  | Study of phyllosphere and rhizosphere microflora                                            | 01 |
| 8.  | Measurement of microorganisms                                                               | 01 |
| 9.  | Preparation of culture media                                                                | 01 |
| 10. | Isolation and purification of N-fixers; Isolation and purification of Nutrient solubilizers | 02 |
| 11. | Isolation and purification of Nutrient solubilizers                                         | 02 |
| 12. | Isolation and purification of Endophytes.                                                   | 02 |

### Suggested readings

1. Pelczar, M.J., Chan, E.C.S. and Kreig, N.R. 2002. Microbiology. 5th Edition, Tata McGraw-Hill, New Delhi.
2. Rangaswami, G. and Bagyaraj, D. J. 2005. Agricultural Microbiology. Prentice-Hall of India Pvt. Ltd., New Delhi.
3. Mukherjee, N. and Ghosh, T. 2004. Agricultural Microbiology. Kalyani Publishers, Calcutta
4. Dubey, H.C. 2007. A Textbook of Fungi, Bacteria and Viruses. Vikas Publishing House Ltd., New Delhi – 10014
5. Salyers, A. A. and Whitt, D. D. 2001. Microbiology: diversity, disease, and the environment. Fitzgerald Science Press, Inc.
6. Prescott, L. M. 2002. Microbiology 5th Edition. McGraw-Hill Inc, US

|                 |                                          |               |
|-----------------|------------------------------------------|---------------|
| <b>STAT-321</b> | <b>Basic and Applied Agri-Statistics</b> | <b>3(2+1)</b> |
|-----------------|------------------------------------------|---------------|

**Objectives:** To provide an idea on Statistical concepts of both descriptive and inference Statistics which will be useful to do statistical analysis.

### Theory

Topic Introduction to statistics and its applications in agriculture. Types of datas, classification of data. Diagrammatic presentation of data. Measures of central tendency, different types of measure. Measures of dispersion, skewness and Kurtosis. Pooled mean & S.D. Probability addition and multiplication laws without proof, simple problems based on probability. Relationship between mean median and mode for symmetrical and shrwed distribution, normal distribution, normal probability curve. Correlation Coefficient with types of correlation coefficient & properties. Regression coefficients, properties of regression roefficients. Tests of significance, definition, one tailed and two tailed tests, test statistic, one sample, two Sample. Paired t-test with examples, F-test for variance. Sampling theory, All definitions are related to sampling theory, Sampling vs census, Simple random sampling with & without replacement. Completely Randomized Design (CRD) of Experimental Designs, Terminologies and ANOVA

and Experimental Designs, Definition of ANOVA, Assignable and Non-assignable Factors, Analysis of One-way Classified Data, Basic Examples.

| SN. | Lecture Schedule Theory                                                                                                                                                                                                       | No. of Lecture |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1.  | Topic Introduction to Statistics and its Applications in Agriculture                                                                                                                                                          | 1              |
| 2.  | Types of Datas, Classification of Data                                                                                                                                                                                        | 2              |
| 3.  | Diagrammatic Presentation of Data                                                                                                                                                                                             | 2              |
| 4.  | Measures of Central Tendency, Different Types of Measure                                                                                                                                                                      | 2              |
| 5.  | Measures of Dispersion                                                                                                                                                                                                        | 3              |
| 6.  | Skewness and Kurtosis                                                                                                                                                                                                         | 1              |
| 7.  | Pooled Mean & S.D                                                                                                                                                                                                             | 1              |
| 8.  | Probability Addition and Multiplication Laws without proof                                                                                                                                                                    | 1              |
| 9.  | Simple Problems based on Probability                                                                                                                                                                                          | 2              |
| 10. | Relationship between Mean Median and Mode for Symmetrical and Skewed Distribution                                                                                                                                             | 1              |
| 11. | Normal Distribution. Normal Probability Curve                                                                                                                                                                                 | 2              |
| 12. | Correlation Coefficient with Types of Correlation Coefficient & Properties                                                                                                                                                    | 2              |
| 13. | Regression Coefficients. Properties of Regression Coefficients                                                                                                                                                                | 2              |
| 14. | Tests of Significance, Definition, One Tailed and Two Tailed Tests, Test Statistic, One Sample, Two Sample                                                                                                                    | 2              |
| 15. | Paired t-test with Examples                                                                                                                                                                                                   | 2              |
| 16. | F-test for Variance                                                                                                                                                                                                           | 2              |
| 17. | Sampling Theory, All definitions are related to sampling theory                                                                                                                                                               | 1              |
| 18. | Sampling vs census                                                                                                                                                                                                            | 1              |
| 19. | Simple random sampling with & without replacement                                                                                                                                                                             | 3              |
| 20. | Completely Randomized Design (CRD) of Experimental Designs, Terminologies and ANOVA and Experimental Designs, Definition of ANOVA, Assignable and Non-assignable Factors, Analysis of One-way Classified Data, Basic Examples | 3              |

**Practical:**

| SN. | Lecture Schedule Theory                                                     | No. of Lecture |
|-----|-----------------------------------------------------------------------------|----------------|
| 1.  | Graphical representation of data                                            | 1              |
| 2.  | Measures of Central Tendency (Ungrouped data) with Calculation of Quartiles | 1              |

|     |                                                                                                  |   |
|-----|--------------------------------------------------------------------------------------------------|---|
| 3.  | Deciles & Percentiles                                                                            |   |
| 4.  | Measures of Central Tendency (Grouped data) with Calculation of Quartiles, Deciles & Percentiles | 1 |
| 5.  | Measures of Dispersion (Ungrouped Data).                                                         | 1 |
| 6.  | Moments                                                                                          | 1 |
| 7.  | Measures of Skewness & Kurtosis (Ungrouped Data)                                                 | 1 |
| 8.  | Moments, Measures of Skewness & Kurtosis (Grouped Data)                                          | 1 |
| 9.  | Correlation and Regression analysis                                                              | 1 |
| 10. | Application of one sample t-test                                                                 | 1 |
| 11. | Application of two sample Fisher's t-test                                                        | 1 |
| 12. | Paired t-test                                                                                    | 1 |
| 13. | Completely randomized design (CRD)                                                               | 1 |
| 14. | Z-test one sample                                                                                | 1 |
| 15. | Z-test two sample                                                                                | 1 |
| 16. | Selection of random sample using simple random sampling                                          | 1 |

**Suggested readings:**

1. Fundamentals of Statistics by D.N. Elhance, Kitab Mahal Publishers
2. Fundamentals of Applied Statistics by S.C.Gupta and V K. Kapoor, Sultan Chand and Sons
- 3 Basic Statistics by B.1.. Agarwal, New Age International Publishers
4. Agricultural Statistics by S.P.Singh and RPS.Verma Rama Publishing House
5. Agriculture and Applied Statistics-I by P.K.Sahu, Kalyani Publishers.
6. Agriculture and Applied Statistics-II by P.K. Sahu and A. K. Dos, Kalyani Publishers